Seminar 6

PROBLEME TRANSFORMAREA FOURIER

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1. Să se calculeze transformata Fourier a semnalului dreptunghiular $f(t) = A(u(t-a) - u(t-b))$ cu $A, a, b \in \mathbb{R}$, $a < b$. Apoi să se arate că $\lim\limits_{|\omega|\to\infty} F(\omega) = 0$.

   R: $F(\omega) = -\frac{A}{\mathrm{i}\omega}(\mathrm{e}^{-\mathrm{i}\omega b} - \mathrm{e}^{-\mathrm{i}\omega a})$

2. Să se calculeze transformata Fourier a impulsului triunghiular de lungime $2T$

$$q_T(t) = \begin{cases} 1 - \frac{|t|}{T}, & \text{dacă } |t| \le T \\ 0, & \text{dacă } |t| > T \end{cases}$$

   R: $F(\omega) = \frac{4\sin^2\frac{T\omega}{2}}{T\omega^2}$

3. Să se calculeze transformata Fourier a funcţiei

$$f(t) = \frac{1}{2}|t-1| - |t| + \frac{1}{2}|t+1|.$$

   R: $F(\omega) = \frac{2}{\omega^2}(1 - \cos\omega)$

4. Să se calculeze transformata Fourier a funcţiei

$$f(t) = \begin{cases} \mathrm{e}^{-3t}, & \text{dacă } t > 0 \\ 0, & \text{dacă } t < 0 \end{cases}$$

   R: $F(\omega) = \frac{1}{3+\mathrm{i}\omega}$

   Să se rezolve ecuaţiile integrale:

5. $$\int_0^\infty x(u)\cos tu du = \begin{cases} 1-t, & \text{dacă } t \in (0,1) \\ 0, & \text{dacă } t \in [1,\infty) \end{cases}$$

   R: $x(u) = \frac{2}{\pi}\int_0^1 (1-t)\cos tu dt$

6. $\int_0^\infty \varphi(\alpha)\sin\alpha x d\alpha = \mathrm{e}^{-x}, x > 0$

   R: $\varphi(\alpha) = \frac{2}{\pi}\cdot\frac{\alpha}{1+\alpha^2}$

7. $\int_0^\infty f(t)\cos\omega t dt = \frac{1}{(1+\omega^2)^2}$

   R: $f(t) = \frac{\mathrm{e}^{-t}}{2}(t+1)$

Rezolvări

(1) Transformata Fourier a semnalului dreptunghiular

$f(t) = A(u(t-a) - u(t-b))$ cu $A, a, b \in \mathbb{R}$, $a < b$

$$f(t) = \begin{cases} 0 & t < a \\ A & a < t < b \\ 0 & t > b \end{cases}$$

$$\mathcal{F}[f(t)](\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt = \int_{a}^{b} A \cdot e^{-i\omega t} dt =$$

$$= \frac{A}{-i\omega t} \cdot e^{-i\omega t} \Big|_a^b = \frac{A}{-i\omega t} \left(e^{-i\omega b} - e^{-i\omega a} \right)$$

$$\lim_{|\omega| \to \infty} |F(\omega)| = \lim_{|\omega| \to +\infty} \frac{|A|}{|\omega|} \cdot |e^{-\omega b} - e^{-i\omega a}| \leq$$

$$\leq \lim_{|\omega| \to \infty} \frac{|A|}{|\omega|} \cdot (\underbrace{|e^{-i\omega b}|}_{\downarrow 1} + \underbrace{|e^{-i\omega a}|}_{\downarrow 1}) = 0 \Rightarrow$$

$$\lim_{|\omega| \to +\infty} |F(\omega)| = 0$$

(4) Transformata Fourier

$$f(t) = \begin{cases} e^{-3t}, & t > 0 \\ 0, & t < 0 \end{cases}$$

$$\mathcal{F}[f(t)](\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{i\omega t} dt = \int_{0}^{\infty} e^{-3t} \cdot e^{-i\omega t} dt$$

$$= \int_0^{\infty} e^{-t(3+i\omega)}\,dt = \lim_{\alpha\to+\infty} \int_0^{\alpha} e^{-t(3+i\omega)}\,dt =$$

$$\lim_{\alpha\to+\infty} e^{-t(3+i\omega)} \cdot \frac{1}{3+i\omega}\Big|_0^{\alpha} = e^{-0\cdot(3+i\omega)} \cdot \frac{1}{3+i\omega} = \frac{1}{3+i\omega}$$

$$\lim_{\alpha\to+\infty} e^{-3\alpha} \cdot e^{-\alpha i\omega} = \lim_{\alpha\to+\infty} e^{-3\alpha} \cdot \left(\cos(-\alpha\omega) + i\sin(-\alpha\omega)\right)$$

$$= \lim_{\alpha\to+\infty} \underbrace{e^{-3\alpha}}_{\downarrow 0} \underbrace{\left(\cos(\alpha\omega) - i\sin(\alpha\omega)\right)}_{\text{mărg}} = 0$$

$\Rightarrow$ Transformata Fourier a funcției f este $\frac{1}{3+i\omega}$

Ecuația integrală

(5.) $\int_0^{\infty} \underbrace{\varphi(\alpha)\cdot\sin\alpha x\,d\alpha}_{f(x)} = e^{-x}, \quad x>0$

Notăm membrul drept cu $f(x)$ și-l prelungim prin imparitate pe $\mathbb{R}$ ca și funcția $\varphi(\alpha)$.

$$\int_{-\infty}^{\infty} \varphi(\alpha)\cdot\cos\alpha x\,d\alpha = 0 \qquad \int_{-\infty}^{\infty} \varphi(\alpha)\cdot\sin\alpha x\,dx = 2\int_0^{\infty} \varphi(\alpha)\cdot\sin\alpha x\,dx$$

$$= 2f(x)$$

$$\int_{-\infty}^{\infty} \varphi(\alpha)\cdot e^{-i\alpha x}\,d\alpha = -2i f(x)$$

Dar $\int_{-\infty}^{\infty} \varphi(\alpha)\cdot e^{-i\alpha x}$ reprezintă transformarea Fourier a funcției φ $\rightarrow$ din formula Fourier de inversare:

$$\varphi(\alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} -2i f(x) e^{ix\alpha}\, dx = \frac{-i}{\pi}\Big[\int_{-\infty}^{\infty} \underbrace{f(x)\cdot \cos x\alpha\, dx}_{=0}$$

$$+ i \int_{-\infty}^{\infty} f(x) \sin x\alpha\, dx\Big) = \frac{2}{\pi}\int_{0}^{\infty} f(x)\cdot \sin\alpha x\, dx =$$

$$= \frac{2}{\pi}\int_{0}^{\infty} e^{-x}\cdot \sin\alpha x\, dx = \frac{2}{\pi}\cdot\frac{\alpha}{\alpha^2+1}$$